

Praveen Kumar .A
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PROFESSIONAL SUMMARY

- Working as Technical Lead Engineer in Einfochips(Arrow Electronics) from July 2023 to till date.
- Working as Lead Engineer in Tech Mahindra Cerium systems from June 2022 to till date.
- Worked as a Sr Engineer - Sys Software Design & Deve in Sasken Technologies Ltd from Feb 2019 to May 2022.
- Worked as Linux Driver Developer in Sigma Computing Solutions from NOV 2015 to FEB 2019.
- 8+ years hand on experience in Device driver development and BSP.
- Programming in C under Linux.

PROFESSIONAL EXPERIENCE

- **Current Company** : eInfochips Private Limited, Bangalore, India.
- **Previous Companies** : Tech Mahindra Cerium Systems Ltd, Bangalore, India.
Sasken Technologies Ltd, Bangalore, India,
Sigma Computing Solutions, Hyderabad, India.
- **Job Title** : Lead Engineer
- **OS used** : Windows, UNIX/LINUX (Ubuntu).
- **Period** : July 2023 to till to date.

TECHNICAL SKILLS

- Languages** : C, Embedded C.
- Technologies** : Linux 3.18, 4.14,4.4, Device Tree.
- Protocols** : SPI, I2C and UART.
- Operating Systems** : Windows, UNIX/LINUX (Ubuntu).
- Debugging Tools** : GCC and,Git, Vim editor and minicom.
- Other Skills** : Working on PnP (Power and Performance),
Worked on Linux Device driver development.
Developing & testing Linux BSP for TI's AM335x,Qualcomm SOC's, Board bring up.

PROFESSIONAL SKILLS:

- Understanding PnP measurement and analysis on windows platform.
- Understanding in programming and debugging C Programming in Linux Environment.
- Experience in Linux Device Drivers & Linux BSP.
- Good exposure to Linux OS, git, patch merging, porting patches.
- Exposure to AM335x, iMX6 and Qualcomm SOC's.
- Worked with Different Buildsystem's (Buildroot, Yocto).
- Proficient in Building Toolchain and BSP (Board Support Package).
- Proficient in C Programming.

PROJECTS

Project 1 : Audio amplifier for car infotainment (Bose)

- Testing the amplifier for different audio modes
- Using Diagnostic tool for testing (PIDT- Porsche car tool)
- Setting up complete test bench with all tools (Optolyzers for MOST)
- Flashing images and checking for Sanity testing.
- Executing MOST commands in optolyzer and observing the behavior of amplifier
- Implemented Temperature Sensor driver update.
- Debugging on temperature sensor issues

Controller : AT32UC3C0512C

OS : OSEK RTOS

Protocol : MOST, 12C, SPI

Language : C

Tools : AVR Debugger, MOST Optolyser, Picoscope, Serial Downloader

Role :

- Technical Lead
- Team member

Project 2 : PnP Power measurement for Intel DTT windows platform (Intel)

1. Setting up new NI DAQ with new RVP.
2. New RVP bring up.
3. Creating header mapping and config file for Power Measurement.
4. Execution of Power KPIs and analysis of CPU PCH Power.
5. Executed some performance KPIs (Like 3D Mark etc.).
6. DTT tuning for Intel RVP and Formfactors.
7. DTT policies execution.

Responsibilities :

- Capture power values on PACS for different KPIs with and without DTT Policies
- Supporting team members for project.

Role :

- Team Lead

Duration : October 2022 to till date.

Project 1: Automotive Telematics (Qualcomm)

1. Yocto build system
2. Field issue debugging
3. Customization to yocto recipes
4. Device driver implementation
5. Security vulnerability fixing

Responsibilities:

- CVE Analysis, Bug fixing and mainlining the changes.

Duration: March 2021 – May 2022

Project 3: BSP Board bring up ADAS

Worked on Board bring up for ADAS project on TDA2SX (TI processor).

1. Worked on Temperature sensor.
 - a. Added the platform data in the device tree.
 - b. Reading the temperature sensor values from the root file system
2. Worked on Supercapacitor
Read the required registers from the root file system by using i2c-tools.
3. EMMY-W161 (u-Blox) WiFi/BT driver
 - a. Get the source code from u-Blox and cross-compiled for the TDA2SX Ti platform.
 - b. Generate the WIFI and BT driver modules and load the drivers and make the WiFi and BT functional.
4. Verified the schematics of the custom board and find the issues and discuss with the hardware team and solve the bugs in the hardware.

Responsibilities:

- Added pin-mux support in Linux
- Added platform data in the device tree (platform data : intr config , reset, clocks configs etc)
- Schematics review

Duration: March 2019 – March 2021

Project 4: Development of I2C based Accelerometer Sensor (ADXL345) Driver for AM335x SOC

1. This project objective is to integrate the Accelerometer Sensor to AM335x SOC, it supports I2C communication protocol.
2. Build for porting on to the target board which uses the kernel 3.14. Ported Linux on to the ARM based development board. In this process I came to know modifying the ADXL driver code and adding it to the device tree source into the kernel source.

Roles & Responsibilities:

- Understood the basic functionality of Accelerometer.
- Understood the Frame format and working of I2C Protocol. Enabled I2C MUX configuration and ADXL driver in kernel.
- Enable I2C Communication protocol in Board specific files in Linux kernel source code Customized ADXL I2C
- client driver in kernel source code. Develop Accelerometer testing application and tested on target board.

Project Duration : March 2018 to Dec 2018

Project 5: HMI based POS

Description:

1. Customize Kernel Driver for Focaltech (FT5x06) M09 firmware with Custom
2. This Project is based on HMI based Device which is having TI's AM335x SOC , with 3.5 TFT Colour LCD display with Touch Screen.
3. FT5x06 is 8 bit low power touch sensor with multi touch capability.

Roles and Responsibilities:

- Modify the kernel driver based on update firmware.
- Added 5 touch feature in Driver.

- Added driver support of SOC.
- Understand INPUT subsystem & I2C subsystem.
- Bug fixes.
- Study the Data sheet.

Project Duration :DEC 2016 to JAN 2018

Academia:

2012	PG Diploma from ISM in Embedded System Design	ISM Hyderabad
2009	B. Tech (Electronics & Communication)	NECG,JNTUA
2001	XII (Intermediate)	Board of Intermediate, A.P
1999	X (10 th Class)	Board of Secondary Education A.P.

Personal Minutiae:

NAME	: PRAVEEN KUMAR A
FATHER'S NAME	: A .LAZARU
DATE OF BIRTH	: 19-08-84
LANGUAGES KNOWN	: TELUGU,ENGLISH
MARRITAL STATUS	: Married
PAN NO	: ATXPA4262E
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I hereby declare that the information provided above is true to my knowledge.

Praveen Kumar Akkelagunta